

Practice: Identifying Population, Sample, and Sample Size

For each of the following, identify:

- a) The population
- b) The sample
- c) The sample size

1. Clothing Store Survey

A total of 684 customers responded to a survey that a clothing retailer emailed to 2,500 customers who had made a purchase during the previous six months. Of the customers who responded, 512 said they were satisfied with the return policy. The retailer wants to estimate how satisfied all of its customers are with that policy.

2. College Students

A college randomly selected 600 of its 8,200 undergraduate students and sent them a questionnaire, hoping to estimate the average number of hours its undergraduates work at paid jobs each week. The average reported work time was 14.6 hours per week among the 427 students who returned completed questionnaires.

3. Supermarket Customers

Among the 1,054 customers who agreed to answer a question about self-checkout, 632 said they use it regularly. Researchers had approached 1,200 customers leaving 15 stores, and 146 of those customers declined to participate. The supermarket chain, which has 74 stores, wants to know what percentage of its customers regularly use self-checkout.

4. Streaming Service

A streaming company sent survey invitations to 10,000 randomly selected subscribers in order to estimate the percentage of its subscribers who watch at least one documentary each month. Of the 2,318 subscribers who completed the survey, 927 reported watching at least one documentary each month. The company has approximately 12 million subscribers.

5. Department Stores

There are approximately 4,600 department-store locations in the United States, and a researcher wants to estimate their average annual sales. Managers at 218 of the 300 stores the researcher contacted provided annual sales information, and the combined annual sales of those 218 stores were \$3.7 billion.